

A Safe and Reliable MultiPlanetary Exchange System

Abstract

We present a prefunded design for trading equities and a capped futures contract across the nine settlements of the MultiPlanetary Exchange System. Each settlement runs one exchange. To trade remotely, a client first transfers assets to an account at the market’s exchange over the backbone. The market executes only against balances already on its ledger. In-transit value is never spendable, so lost, delayed or duplicate packets can increase latency but cannot create or destroy money or shares.

We evaluate the design with a discrete-event simulator of the network in the brief. Without packet loss, an Earth client buys 1,000 Ares shares at Mars and receives them at home after 3.07 hours. With random loss, the median is 3.07 hours, the 99th percentile is 7.70 hours, and all 2,000 simulated runs complete. The 300-hour capped futures contract settles correctly on both rising and falling price paths, with both payouts spendable at home after 328.17 hours. All nine settlements can access every implemented product. Neptune is slowest: a share purchase completes within 72 hours with probability 0.47 and within 168 hours with probability 0.97.

1 Introduction

We design and test a financial market system for nine settlements, from Mercury to Neptune. We implement equities, a capped futures contract, and their clearing and settlement. Section 2.3 describes how other products, including options and currencies, could extend the same framework. All inter-settlement communication passes through gateways. Exchange operators may use the more reliable backbone for official coordination; clients use local access or the lossy direct service. Messages can take minutes to hours, packets can be lost, and solar exclusion can block paths for weeks. No exchange can therefore assume that it knows another exchange’s current state.

The design does not require that knowledge. Each balance has exactly one writer, and an exchange acts only on assets already recorded in its own ledger. Inter-exchange transfers are credited exactly once. Packet loss, delay and duplication can therefore slow service but cannot change conservation or allow the same asset to be spent twice. Unlike a distributed commit protocol [4], an exchange never waits for another exchange to agree before changing its own ledger.

Section 2 describes the architecture, Section 3 the operating rules, and Section 4 the guarantees and costs. Section 5 reports risks found in testing, Section 6 covers long-term operation, and Section 7 discusses deployment effects. The evidence appendix contains the traces, calculations and simulation results supporting the numerical claims. Unless stated otherwise, all results use epoch 2026-09-22 00:00 TDB.

2 Architecture

2.1 Institutions

The system has one exchange at each settlement. Each exchange keeps its customer accounts in its own ledger and is the sole writer of those balances.

Any exchange may host a market. In our scenarios, the stock markets are colocated with their registries and initial sellers: Ares Habitat at Mars, Belt Works at Ceres, and Terra Fabrication at Earth. Mars also hosts the capped futures contract; it is well connected to the inner system (Appendix S3), and Bob, the price source, is local.

2.2 Accounts

Each client has one registered home settlement. A client may also hold a trading account at any of the other eight exchanges. Remote accounts start empty and receive assets only through explicit transfers via the home exchange. Opening a remote account does not create money, a new identity or any additional direct-service quota. Table 1 summarizes which institution may change each part of the shared state.

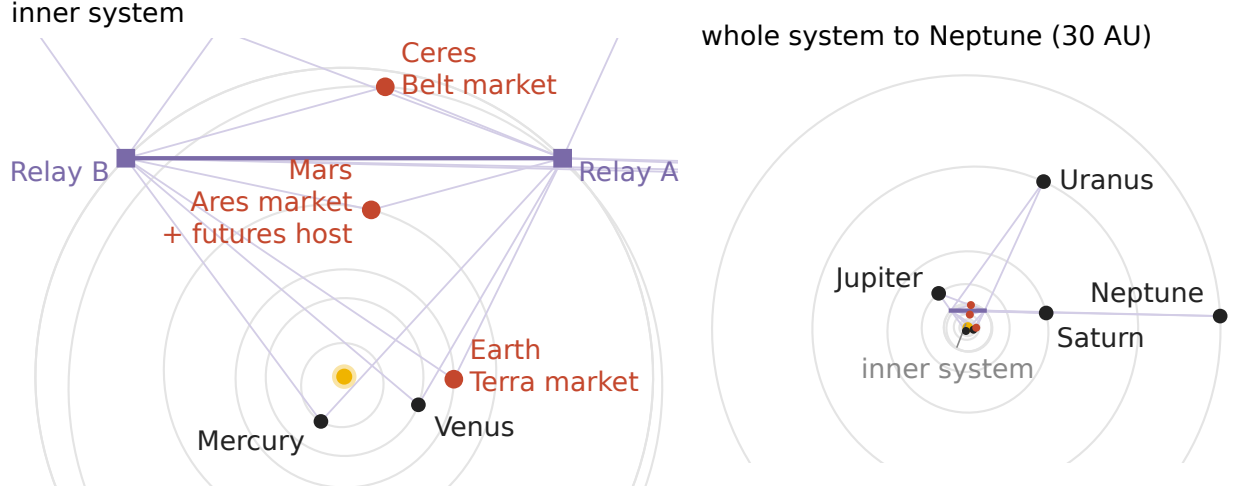


Figure 1: A simplified chart of the nine settlements and two relays. Purple lines represent the 19 backbone links, and red markers are settlements that host a market. Actual flights do not exactly follow straight lines between the positions shown, and a link is closed whenever its path passes within 0.1 AU of the Sun.

Table 1: Ownership of shared state

Shared Information	Governing Institution
Available balances, order holds and contract escrow	The exchange that holds those balances
Order book and fills	The asset's market exchange
Debit and terms of an outgoing transfer	The sending exchange
Credit of an incoming transfer	The named receiving exchange, exactly once
Share registry (how many shares each exchange holds)	The asset's registry/market exchange
Contract positions, fixing and payout allocation	The contract host

Every run starts from Table 2; only clients' home accounts initially hold assets.

Table 2: Opening balances

Account	Alice	Bob	Cara	Dax	Eve	Fin	Total
Home	Earth	Mars	Ceres	Neptune	Uranus	Earth	5 settlements
NeoDollars	\$150,000	\$50,000	\$100,000	\$100,000	\$50,000	\$50,000	\$500,000
Ares	—	3,000	—	—	—	—	3,000
Belt	—	—	1,000	—	—	—	1,000
Terra	—	—	—	—	—	1,000	1,000

2.3 Products and scope

Table 3 lists each product in scope and marks it as implemented or an extension. In either case, an exchange may act only on assets already in its ledger, and each balance has exactly one writer.

Table 3: Status of all products in the brief’s scope

Product	Status	Treatment in this framework
Equities (Ares, Belt, Terra)	Implemented	They can be bought and sold in units of one share through limit orders at the market exchange (R2), fully prefunded with no end of life.
Capped futures	Implemented	Size Q per index point, strike $K = 100$, collar $[60, 140]$, 300 h term, hosted at Mars, priced by Bob. The long gains when the fixing price $P > 100$, and the short gains when $P < 100$. Because P is capped to $[60, 140]$, each side’s maximum loss is 40Q, so each side escrows 40Q before opening. The contract ends at fixing with automatic payouts.
Clearing and settlement	Implemented	No central counterparty. Delivery versus payment in one ledger write at the host (R2); transfers between exchanges by R1.
Capped options	Extension	Writer escrows the maximum payout and the buyer prefunds the premium; settles like R4.
Short positions	Extension	The bounded futures short is implemented. Borrowed-share shorts need borrow, return and buy-in rules and are not supported.
Bonds and lending	Extension	A loan creates a claim and a liability, not cash. Guaranteed repayment needs escrowed backing; volatile collateral cannot guarantee it.
Currencies	Extension	A unit issued by one exchange and fully backed by escrowed NeoDollars, redeemable only there.

2.4 Example: an Earth client buys Mars shares

A cross-settlement trade proceeds in five steps. Figure 2 shows these steps for scenario S1a.

1. *Funding.* Alice instructs her local exchange, the Earth exchange, to move \$100,000 to her Mars account. Earth debits her balance immediately and sends a **TRANSFER** record over the backbone. While the record is in flight, the money still belongs to Alice, but nobody can spend it.
2. *Import.* Mars accepts the transfer exactly once, credits Alice’s Mars account and sends a **RECEIVED** record back to Earth.
3. *Order.* When Earth has the receipt, Alice knows that her money is at Mars. She then sends a signed order to Mars by the direct service. She sends several copies because direct packets are often lost.
4. *Trade.* Mars holds \$100,000 of Alice’s cash. It matches her order exactly once against Bob’s resting sell order and exchanges cash for shares in a single atomic step. Bob can spend the proceeds at Mars immediately.
5. *Delivery.* Alice asked for home delivery, so the same atomic step also creates a transfer of her 1,000 shares to Earth. For Alice the trade is complete when Earth credits the shares.

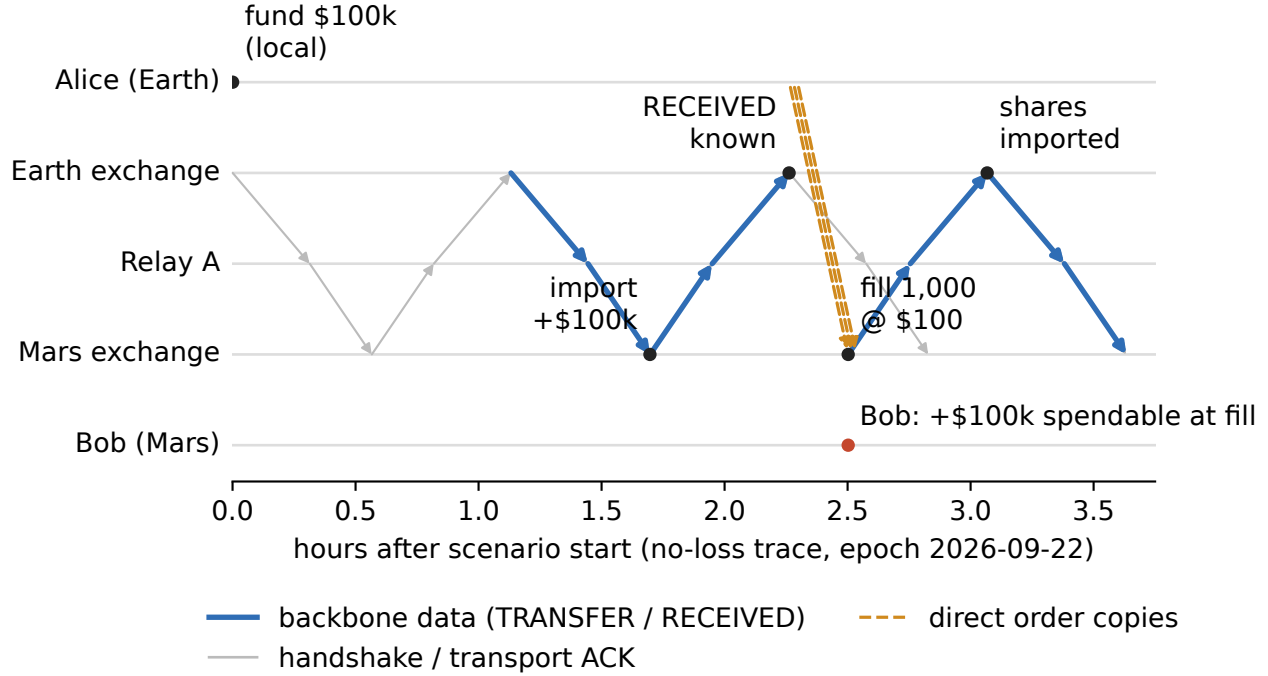


Figure 2: The S1a trade. Bold arrows carry money or shares. Grey arrows are handshake and acknowledgement packets required by the network. Dashed arrows are Alice’s three direct copies of her order. The trade is complete at 3.067 h.

3 Operating rules

Each rule below states who acts, what information they may rely on and what happens when a message is lost or late. A change is *durable* if it survives a crash or reset. Every change of financial state described here is a single atomic, durable write to one exchange’s ledger.

3.1 R1: Moving assets between exchanges

A TRANSFER record has a globally unique identifier (source exchange and export number), owner, asset, amount, destination and purpose. Its terms never change.

1. The source checks the owner’s *available* balance and, in the same atomic write, debits it and creates the transfer. The value is then “in transit” and cannot be spent elsewhere.
2. The destination credits the transfer the first time it sees the identifier and keeps a durable duplicate record (compacted in Section 6). Later copies only cause another RECEIVED.
3. The source marks the transfer delivered when RECEIVED arrives.

A missing RECEIVED never causes a refund: the destination may already have credited the transfer and only the confirmation may be lost. The network first uses its hop and endpoint retries; if those fail, the source resubmits the same TRANSFER after 24, 48 and 96 hours and then every 168 hours. Reusing the identifier makes the retries idempotent. If the destination account is closed, the asset enters a restricted, non-spendable balance and is returned in a new transfer.

Share transfers always pass through the stock’s registry exchange. Shares may move between the registry and a custodian exchange, but never directly between two custodians. The registry can therefore reconcile how many shares each exchange holds.

3.2 R2: Orders, holds and matching

An order specifies its owner, unique identifier, asset, side, integer quantity, limit price, time-in-force (good-till-cancelled or immediate-or-cancel), expiry no more than 30 days away, and disposition. The disposition

says whether proceeds stay at the market or are sent home automatically.

The market accepts an order only if the owner’s available balance covers it: quantity times limit price for a buy, or the shares for a sell. Those assets move into a hold that nothing else can use. An uncovered order is rejected permanently; after funding, the client must submit a new order with a new identifier.

Orders match by price-time priority at the resting order’s price, and self-trades are skipped. Each fill is atomic: the buyer’s cash hold and seller’s share hold are reduced, the seller receives the trade price, and the buyer receives the shares plus any price improvement. If the buyer requested home delivery, the same write creates the transfer home. This is delivery versus payment within one ledger [3]. An unmatched GTC order rests until filled, cancelled or expired; an IOC order immediately releases any unfilled remainder. A cancellation releases only the unfilled portion. If a cancellation arrives before its order, the market records it and rejects the later order. Orders cannot be amended; they must be cancelled and replaced.

Listings and quotes. Ares trades only at Mars, Belt only at Ceres and Terra only at Earth, in whole shares at whole-NeoDollar prices. Every exchange holds this listing table, so a client can learn locally whether a stock exists and where it trades before funding. Listing changes are official records sent over the backbone with a future effective time; none occurs in our scenarios. Each market keeps the authoritative book for its stock. Quotes from R5 are informational snapshots and may be stale. A client therefore submits a limit price rather than relying on a quote for execution, and the system has no market orders.

Accepted orders have states `RESTING`, `PARTIALLY-FILLED`, `FILLED`, `CANCELLED` or `EXPIRED`; rejected orders have state `REJECTED`. A remote client learns material state changes through the R5 status mechanism. These reports are informational only: losing or delaying a report cannot change the order or its financial effect.

3.3 R3: Withdrawals and account closure

A `WITHDRAW` instruction names an amount or `ALL`; `ALL` is evaluated once when the host first processes it, and duplicates return the same transfer. Withdrawals use only available balance and do not cancel orders or touch margin. A `CLOSE` instruction rejects new orders, cancels open ones and withdraws the free balance. Escrow remains at the host until the contract settles, after which payouts are sent home automatically. Only the exchange holding an asset can send it back.

3.4 R4: Implemented capped futures and margin rule

We implement one capped cash-settled futures contract as the tested price-dependent product. Other derivatives may use different payoff functions, but any supported contract must have a finite maximum loss for each side and fully escrow that amount before opening.

For the implemented contract, the long and short agree on a size Q in NeoDollars per index point, strike $K = 100$, and collar $[60, 140]$. The long’s payoff is

$$Q \times (\min(\max(P, 60), 140) - 100),$$

where P is the fixing price; the short receives the negative.

Margin rule. Each side escrows its maximum possible loss before the contract opens. Here that amount is $40Q$ per side, held at Mars until fixing. There is no variation margin, margin call, netting or share collateral. Because escrow covers the maximum obligation, no price path can create an unfunded liability.

Each party first funds its Mars account (R1) and then sends a contract instruction containing the full terms. Mars moves that party’s margin from available cash into a hold. When both holds exist, the contract opens atomically and matures 300 hours later. If both margins are not held by the opening deadline, 168 hours after the scenario starts, the contract is rejected and each hold returns to available balance. A client that has not received `OPEN` may withdraw its available balance. At fixing, payout transfers home are created in the same atomic write.

Bob is the named index source at Mars. He publishes one signed observation locally at 60, 120, 180, 240 and 300 hours after opening. Mars fixes the contract 24 hours after maturity using observation 5. If it is missing, or conflicting copies exist, Mars uses the latest earlier valid observation, or 100 if none exists. Observations arriving after fixing are ignored.

3.5 R5: Communication services

Table 4 assigns each message type to a communication service. The nine exchanges receive 66 of the 600 daily backbone originations each, using 594 in total and avoiding a shared counter. SYN packets, first transmissions, **RECEIVED** records and application resubmissions count against this quota; automatic retries and ACKs do not, although packet totals include every launch. Client instructions do not use the backbone, and exchanges do not use the direct service for official coordination.

Quota classes. A data packet carrying any record other than **RECEIVED** is *routine* and may be originated only while the exchange has used fewer than 44 originations in the previous 24 hours. SYN packets, receipt-only packets and application resubmissions may use all 66, leaving at least 22 originations per exchange for recovery.

Packet sizes. Every application record has an 88-byte common header containing type, length, identifier, creation time and issuer signature. **TRANSFER** is 176 bytes, **RECEIVED** 120 bytes, a quote request 120 bytes, a market snapshot 152 bytes, and a client instruction 128–256 bytes. A network packet has the brief’s 64-byte header and at most 960 payload bytes; our 16-byte sub-header leaves 944 bytes for whole records. Records are never split, and a packet closes when the next record would not fit.

Status reports. Remote clients learn material order and contract state through their home exchange. A market or contract host batches **STATUS** entries by destination home exchange and sends them over the backbone; the home exchange reports them to its client by local access. Equity states are **REJECTED**, **RESTING**, **PARTIALLY-FILLED**, **FILLED**, **CANCELLED** and **EXPIRED**; contract states include **HELD**, **OPEN** and **REJECTED**. Status is informational only and does not change the underlying financial state. Backbone delivery makes it usable even when the client’s direct path is blocked. Contract-status traffic is already included in Appendix S1 packet counts; an equity status that can share a packet with an existing transfer or receipt adds no launch, while a standalone status consumes routine backbone capacity.

Market data. Each market can return a 152-byte snapshot containing best bid and ask with sizes, last trade, book version and timestamp. Every exchange keeps the newest snapshot it has received for each stock on a local *board*; at the market itself, the board reflects the live book. A client reads its local board and may issue a **QUOTE** request to its own exchange. If the snapshot is older than the client’s requested freshness, that exchange may request a new snapshot from the market over the backbone, at most once per stock every 6 hours and shared across its clients. The request and reply are routine records.

Quotes are not execution guarantees. They may be stale by the time an order reaches the market, so the client’s limit price determines the worst acceptable execution price. Market data is request-driven and never uses the direct service. In S1a, the quote request fit in Alice’s funding packet and the reply fit with **RECEIVED**, so the scenario’s packet count did not change.

Table 4: Communication service for each message type.

Message	Service	Quota charged
TRANSFER and RECEIVED	Backbone, fixed route of at most 3 links	Sending exchange; up to 66 originations per rolling 24 h
Order, cancel, withdrawal, close, contract instruction	Local access at host; otherwise direct	Client’s 12 direct packets per rolling 24 h, at least 60 s apart
Price observation	Local access at Mars (Bob is local)	None
Listing table, local market-data board and QUOTE instruction	Local access at client’s exchange	None
Quote request and snapshot reply	Backbone between client’s exchange and market; at most one request per stock per 6 h	Request counts against requesting exchange’s routine quota; reply against market’s routine quota
STATUS	Backbone from host to client’s home exchange, then local access	Host exchange’s routine backbone quota; batched by home exchange

3.6 R6: Reference client behavior

Clients are not trusted to enforce financial safety; exchanges enforce balances, holds, identifiers, deadlines and duplicate handling. Our evidence uses a reference client that knows the published ephemeris but not

hidden incidents. Its behavior affects latency and packet use, but not asset safety.

- *Fund first, then instruct.* The client waits until its home exchange records **RECEIVED** for the funding transfer before sending the order or contract instruction, avoiding rejection for insufficient funds.
- *Read available market data.* When preparing a remote stock order, the client reads the listing table and local market-data board. It may request a fresher quote while funding. A stale or missing quote does not block submission; the client chooses its own limit price, which bounds the execution price.
- *Direct delivery.* The client sends the fewest direct copies, 60 seconds apart, needed for at least 99% delivery probability, up to 8 copies. If the direct path is closed, it uses R8 forwarding through the third settlement with the earliest arrival.
- *Fallback.* If no result arrives within twice the round trip plus 6 h (12 h at most for contract instructions), the client resends the same instruction, directly or through R8, within its 12-per-day quota, until the order expiry, opening deadline or 30 days. Released cash stays available at the host for withdrawal. The host processes each identifier at most once, so duplicates have no additional financial effect.

3.7 R7: Recovery

Ledgers, outboxes and identifier records are durable. After a reset, session expiry or failed network delivery, the courier opens a new session and resubmits the same records after 24, 48 and 96 hours and then every 168 hours. It waits out known closures and quota limits and prioritizes recovery traffic. Table 5 summarizes failure handling.

Table 5: Failure handling.

Event	Sender	Receiver	Financial effect
TRANSFER lost or late	Has no RECEIVED and resends the same transfer	Knows nothing; value is not spendable there	Value remains in transit and owned by the client
RECEIVED lost	Resends TRANSFER	Ignores duplicate and resends RECEIVED	None; value is already spendable at destination
Duplicate transfer/order/instruction	—	Returns original result; changed terms are rejected	None
All order copies lost	Resends same order ID after fallback timeout	No order exists	Cash remains available at host
Cancel lost or late	Assumes nothing until order state is known	Order may still fill	Fill stands; cancel releases only unfilled quantity
Quote stale or lost	May request a fresher quote	Keeps newest snapshot by version	None; limit price bounds execution
STATUS lost	Courier resends it	Client may resend original instruction	None
Conflicting price copies	—	Uses latest earlier valid observation	Fixing is final and never revised
Host isolated for 72 h	Remote instructions delayed or forwarded	Local matching and maturity continue	Funded payouts wait in durable outbox
Exchange reset	Open sessions are lost	Ledger and outbox remain durable	None
Forwarder slow or silent	Client retries after fallback timeout	Host sees nothing or a duplicate	None; funds remain at host

3.8 R8: Forwarding through a third exchange

Solar exclusion can block a client’s direct path to the host for weeks while the backbone remains connected. Because client instructions cannot use the backbone, R8 permits forwarding through a third settlement. When the direct path is closed, the client chooses a settlement whose two direct legs are open and whose route gives the earliest arrival.

1. The client wraps its signed instruction in a **FWD** record with attempt number $k = 1, 2, \dots$ and sends it to the third exchange by direct service, using the R6 copy rule for that leg.
2. The third exchange forwards each new pair (instruction identifier, k) once under its own 12-packet rolling 24-hour direct quota, again using the R6 copy rule. Duplicate copies of the same attempt are ignored. The pair is retained until the instruction expires or its opening deadline passes.
3. The host treats the forwarded instruction exactly like a direct one. It verifies the client’s signature, balance and identifier, so direct and forwarded copies of the same instruction are duplicates.

The forwarding exchange gains no authority: it cannot alter the signed instruction, no asset passes through it, and it has no financial obligation. The client learns the result through its home exchange using the R5 status mechanism. If no status arrives before the R6 fallback timer, the client sends attempt $k + 1$ through the best route then available, or directly once the path reopens. Forwarding uses two direct legs and therefore roughly twice as many direct packets. During the 2028 Mars conjunction, Earth cannot reach Mars directly for 53 days; with R8, S1a completes in 4.55 hours (Appendix E5).

4 Guarantees and costs

We distinguish between properties that the *rules* guarantee in every possible run and properties that the *model* (orbits, loss law and quotas) makes likely. The first kind holds for any pattern of packet loss, delay or reordering. The second kind is supported by simulation evidence but is not proven. Table 6 lists both.

Table 6: Guarantees of the design and the conditions they depend on.

	Guarantee	Reason	Condition
G1	The total of NeoDollars and of each share type never changes	Every write debits and credits equal amounts; value in transit is a ledger state	Rules only; checked after every event of every traced run
G2	No asset is used for two purposes at once	Available, held, escrowed and in-transit balances are separate	Rules only
G3	Every futures payout is fully funded	Escrow equals the maximum loss and is posted before opening	Rules only; holds for any price path
G4	Each transfer is credited exactly once	The destination keeps a durable record of every credited identifier	Rules and durable storage
G5	A fill is final and immediately spendable at the host	Both holds exist before the atomic fill	Rules only
G6	A funded transfer eventually arrives	Retries never stop, the backbone is always connected, and admission keeps work within quota	Model; under random loss there is no fixed deadline
G7	Completion times	Measured in simulation	Model; see Table 7

4.1 Service at every settlement

Every implemented product is accessible from every settlement. Clients can fund and withdraw at all three stock markets, trade Ares, Belt and Terra shares, and enter the capped futures at Mars. Only latency and completion probability vary by settlement; Table 7 summarizes both. *Backbone min* is the one-way delay in minutes to the Mars, Ceres and Earth hosts. *Copies*, P represent the direct copies per instruction to Mars and the probability that at least one arrives. The next three columns are for a \$100,000 Ares purchase with home delivery: completion time in hours without loss, the median under random loss, and the probability of completing within 72 hours (1,000 runs each; Appendix S3).

Table 7: Service at each settlement at hour 0.

From	Backbone min	Copies, P	No loss	Median	P(72 h)	Notes
Mercury	42 / 42 / 46	3, 0.997	3.8	3.8	1.00	Direct path to Mars closed from h 22 to h 357; instructions are forwarded (R8)
Venus	37 / 36 / 41	3, 0.997	3.3	3.3	1.00	Closed near Venus–Mars conjunctions; instructions are forwarded (R8)
Earth	34 / 33 / local	3, 0.998	3.1	3.1	1.00	Terra local. Direct path to Mars closed about 53 days every 2.1 years; forwarded (R8)
Mars	local / 30 / 34	local	0.0	0.0	1.00	Ares and futures local; Belt (Ceres) and Terra (Earth) remote
Ceres	30 / local / 33	2, 0.992	2.6	2.6	1.00	Belt local; Ares and futures (Mars) and Terra (Earth) remote
Jupiter	40 / 41 / 51	4, 0.992	3.9	4.0	1.00	4 copies per instruction
Saturn	76 / 75 / 79	7, 0.990	7.6	10.6	1.00	7 copies per instruction
Uranus	155 / 154 / 158	8, 0.885	15.4	26.8	0.99	8 copies; the fallback is often used
Neptune	247 / 246 / 250	8, 0.545	24.7	74.4	0.47	8 copies still all fail 46% of the time

Table 7 uses the Ares purchase; Appendix S3 repeats the test for every implemented product and settlement. A market is local only at its host: Ares and futures at Mars, Belt at Ceres, and Terra at Earth. For remote share purchases, the three products finish within 1.1 hours of one another without loss because client location dominates the delay.

Every futures party must reach Mars before the 168-hour opening deadline. Parties in the inner system almost always do: in all 2,000 simulated runs with random loss, the contract opened (Appendix E2). For a Neptune party (Dax long against Cara), the contract opened before the deadline in 99.7% of runs with random loss. In the other runs it was rejected at the deadline and all margin was returned (Appendix S3).

4.2 Efficiency

The design’s safety guarantees hold in every evaluated scenario: money and shares are conserved, assets cannot be spent twice, and every accepted obligation is fully funded. The remaining tradeoff is efficiency, mainly in communication and capital use.

Communication efficiency is the total number of direct and backbone packets divided by completed transactions. A remote equity purchase in S1a uses 15.7 packets per transaction. Reusing an already funded account is cheaper: S1i completes four fills, a cancellation, a cancel–fill race and a final transfer home at 8.4 packets per transaction. The capped futures contract uses 28.2 packets per transaction because opening, status reporting, fixing and payout require more communication.

Capital efficiency measures how much NeoDollar value must be encumbered to support the value settled. S1a has a ratio of 0.33, while the capped futures contract has a ratio of 0.31. The ratio alone does not show how long capital is unavailable, so we also report asset-hours. In S1d/e, \$160,000 remains encumbered for 324 hours, or 51.84 million dollar-hours. This prolonged capital lockup is the main economic cost of fully prefunding the futures contract.

5 Risks

Table 8 lists weaknesses found through incident search, shifted-epoch runs and outer-settlement tests, ranked by expected harm under our evidence. The most important arise from interactions between protocol rules and orbital geometry.

Table 8: Risks found during testing, ranked by harm and likelihood.

	Risk	Evidence	Treatment
1	A solar conjunction cuts a client off from its host	During the 2028 Mars conjunction the Earth–Mars direct path is closed for 53 days. Earth clients cannot order, cancel or withdraw at Mars. Bob’s 30-day sell order expires first, so the trade does not happen (E5).	Rule R8 forwards instructions through a third exchange (Mercury here): the trade completes in 4.55 h and both futures runs complete.
2	Money left at a host can become unreachable	Withdrawals are client instructions sent by direct service. Money left at Mars during a closure cannot be called home, even though the backbone still works.	Clients fund just before they trade (R6). Home delivery and payouts need no instruction. A withdrawal can be forwarded through a third exchange (R8).
3	Direct loss of 76–91% per copy at the outer settlements	From Neptune, $P(\text{complete within 72 h}) = 0.47$, because 8 copies only succeed 54% of the time (S3).	Up to 8 copies of direct packets are sent, to balance tight quota limits with the high chance of packet loss for outer settlements.
4	Concentration at the host	All futures margin and the price source are at Mars. An isolation of Mars around the fixing delays both payouts by up to 75 h (S2 search).	Fixing and payout are local and automatic. Payouts wait, already funded, in the outbox.
5	A direct path can close after funding but before the next instruction	In an earlier design, Mercury funded Mars shortly before the Mercury–Mars path closed from h 22 to h 357, leaving the client unable to instruct the host directly (S3).	R8 removes the dependence on a continuously open direct path by forwarding instructions through a third settlement.
6	Sessions are silently lost after an exchange reset	The worst of 56,280 simulated incidents (S2) is a Mars reset at h 0.75, just after the handshakes. Mars silently drops the funding transfers, and the senders wait about 100 h for endpoint retries plus a 24 h backoff. The futures complete at 453.79 h instead of 328.17 h. The contract still opens before the deadline, and no money is at risk.	Alternative A1: after a reset, the exchange sends a restart notice to its recent peers. The futures then complete at 329.49 h. A1 is not part of the evaluated design; it is the first change for the next revision.
7	Status reports at scale	If Mars reported to each remote client by direct service, its 12 direct packets per day would cover at most 12 status changes a day for all clients together.	Status travels as batched STATUS records to home exchanges over the backbone (R5), at most one entry per client per 6 h unless the state changes. Clients still rely on fallbacks.
8	Single price source	Bob may fail or publish conflicting copies.	The fixing falls back to the latest valid earlier observation, or 100. This basis risk is disclosed.
9	Trust in the custodian exchange	Prefunding concentrates exposure on the host exchange.	The brief assumes operators follow the rules. Independent audit is future work.

6 Long-term operation

The backbone never disconnects. The two relays are 90° apart on the same circular orbit, and the link between them always passes at least 2 AU from the Sun. A link between a settlement and a relay is blocked only when the settlement is almost directly behind the Sun as seen from that relay, at an angle of at least 159°. This cannot happen for both relays at the same time. Every settlement therefore always has a route of at most three links to every other settlement, except during maintenance and incidents. Our 200-year

scan confirms this: every settlement had a route to every market host at all 1,753,201 hourly samples. The slowest one-way route observed was Mars to Neptune at 315 minutes. Because a route has at most three links, its length is at most $(r_a + R) + 4 + (r_b + R)$ AU at any time, where r_a and r_b are the aphelion distances of the two settlements and $R = 2.83$ AU is the relay radius. With empty queues this bounds every one-way route delay for all time; the largest bound is 358 minutes (Neptune to Ceres). Direct paths, in contrast, do close for weeks at a time during conjunctions, so every promise that depends on the direct service without rerouting is conditional.

Long outages and the backlog afterwards. Only two things have a deadline: orders, which expire after at most 30 days and release only their own hold, and contract opening deadlines, which return the margin. Transfers, receipts and payouts wait in durable outboxes. Sessions that are idle for 7 days are reopened. A packet that reaches the network’s 30-day lifetime is replaced by an application resubmission with the same financial identifier. When a route reopens, each exchange works through its outbox in packets of five transfers, within its 44 routine originations per day. We say a backlog has *cleared* when every transfer is credited at its destination and the source holds every **RECEIVED**. In simulation (Appendix E6), with no competing traffic, 300 transfers from Earth to Mars are credited within 24.6 hours and resolved within 25.1 hours of the route reopening, and in at most 30 hours in 20 runs with random loss. Other work from the same exchange shares the 44 routine originations: 300 transfers to each of the eight peers clear in about 10 days.

Obligations further in the future. A contract stores its maturity and fixing times as absolute times. The fixing is a local computation at the host, and the payout transfer is created in the same write. A maturity several years away therefore needs no message in order to take place.

Capital. New capital can enter only through fees or contributions debited from real accounts and moved by R1. Escrow is released only by fixing or rejection; the system creates no free capital.

Storage and identifiers. An identifier is 12 bytes: a 32-bit principal prefix and a 64-bit counter. At a million identifiers per second one principal’s counter lasts about 585,000 years. Times are stored as signed 64-bit integer milliseconds, which covers about 292 million years in each direction. Our simulator uses floating-point seconds, which can no longer resolve 1 ms after 142,710 years, so a production system must use integer time and extended-precision orbital phase. State is bounded: at most 32 live orders per client per host, 256 per host, and 4,096 export slots per host, each an unresolved transfer or a reservation. A remote client’s contract instruction reserves its payout’s slot and a home-delivery order its delivery’s (one delivery in transit per order). An instruction is refused if the slots it needs would take the total past 3,276 (80%), so every payout and delivery has its slot before it exists. A late top-up to a closed account waits in a restricted balance, owned by the client, until a slot is free, so slots in use never exceed 4,096.

Compaction without losing duplicate protection. A source numbers its transfers to each destination consecutively and sends them in that order. When a transfer is credited, the destination replaces the full record by a *tombstone*: the transfer identifier, the final state **CREDITED** and a 32-byte hash of the terms. A later copy that matches a tombstone only causes another **RECEIVED**, and a copy whose terms do not match the hash is rejected. For each source the destination also keeps a *watermark* W , the highest number such that every transfer up to W has been credited. Tombstones at or below W are dropped, because any identifier at or below W is known to be a duplicate. A source does not send transfer n while any transfer numbered $n - 512$ or lower is unresolved, so every credited transfer above W lies within 512 of it: at most 512 tombstones per source, however long a gap stays open. The source compacts its own export to a tombstone when the **RECEIVED** arrives. Order and instruction identifiers are compacted the same way, and any copy that arrives after the order’s expiry or the contract’s opening deadline is rejected on that ground alone. The result of a withdrawal, funding, cancel or close request is kept for 60 days, and a copy older than 30 days is rejected. Transport packet IDs follow the brief’s own retention rule. A 52-byte closure record diverts late top-ups until the host’s watermark passes the home’s last funding sent before it learned of the closure; a contract tombstone lasts until its instructions expire (Supplement E6; design rules, not simulated). Full records can move to an off-line audit archive; the live ledger needs only balances, open records, tombstones above the watermarks and one watermark per peer. When a limit is reached the exchange refuses new work; an obligation it has accepted already holds the capacity it needs.

Accepted demand. Each exchange can make up to 44 routine backbone originations per day, which is enough for about 10 new funding or withdrawal transfers between exchanges at 4 to 5 originations each, plus 22 originations for recovery. Market data adds one reply per request, at most 4 a day for each exchange and stock (32 a day at a market), and nothing when no one asks. Each client has 12 direct packets per day, which is enough for 1 to 4 remote instructions depending on distance. A forwarding exchange (R8) also has 12 direct packets per day, so it can forward 1 to 4 instructions a day. Work beyond this is queued, and we make no promise about its timing.

Very long horizons. The connectivity argument and the delay bound above depend only on the shapes of the orbits, so they hold at every time in the model. The scanned statistics (availability, closure counts and typical delays) describe only the 200 scanned years. The orbital periods are not commensurate, so the joint configuration of the bodies never repeats exactly, and we do not extend those numbers beyond the scan. Our evidence for later epochs is the runs at +1, +10 and +100 years (Appendix E5). With integer milliseconds and extended-precision phase, nothing in the rules depends on the date. The remaining limits are the identifier space described above and the physical infrastructure, which the brief fixes.

7 Deployment assessment

The baseline omits several real effects that would change latency or loss rates but not the ledger rules that enforce conservation and exclusive use. Table 9 estimates their size.

Table 9: Effects a real deployment would add.

Effect	Estimated size	Source and assumptions	Effect on the guarantees
Earth rotation	About 11.97 h without view per ground station per day	Sidereal day of 23.93 h, one equatorial station, no elevation mask [1]	Delays only, up to 12 h per hop with one station; G1–G5 unchanged.
Doppler shift	About 3.20 MHz at 32 GHz	$f \cdot v/c$ with an assumed line-of-sight speed of 30 km/s [1, 2]	Modems track it; otherwise more loss, which only slows service.
Relativistic clock rates	Mercury vs Earth: 2.3×10^{-8} , or 0.7 s per year	$GM/(rc^2) + v^2/(2c^2)$ at 0.39 AU and 1 AU	Maturity and fixing use TDB; local clocks are corrected.
Ephemeris drift	Not estimated	Kepler elements ignore perturbations between planets	Timetables need a live ephemeris.

References

- [1] NASA, Earth Fact Sheet. <https://nssdc.gsfc.nasa.gov/planetary/factsheet/earthfact.html>
- [2] ESA, Cebreros DSA-2 ground station (Ka-band, 32 GHz).
- [3] CPMI–IOSCO, *Principles for Financial Market Infrastructures* (delivery versus payment), 2012.
- [4] J. Gray and L. Lamport, Consensus on Transaction Commit, *ACM TODS* 31(1), 2006.